Xinyuan HUANG

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Education

The Pennsylvania State University University Park, PA
Department of Civil and Environmental Engineering
Ph.D. in Environmental Engineering Aug 2019 – Mar 2024
Advisor: Dr. Wei Peng; Co-advisor: Dr. Klaus Keller

Stanford University Department of Civil and Environmental Engineering Stanford, CA
M.S. in Environmental Engineering and Science Sep 2015 – Jun 2017

Peking University Beijing, China
B.S. in Chemistry, College of Chemistry and Molecular Engineering Sep 2011 – Jul 2015
B.A. in Economics, National School of Development Sep 2012 – Jul 2015

Professional Experiences

Princeton University, Andlinger Center for Energy and the Environment Princeton, NJ
Postdoctoral Research Associate Apr 2024 – Present

The University of Maryland College Park, MD
Visiting Fellow at Center for Global Sustainability Sep 2023 – Dec 2024

The Joint Global Change Research Institute College Park, MD
Visiting Fellow May 2023 – Aug 2023
Advisor: Dr. Steven Smith

Massachusetts Institute of Technology Cambridge, MA
Research Associate at Sloan School of Management Jul 2018 – May 2019

S.F. Express Co. Ltd Shenzhen, China
Data Scientist Jul 2017 – May 2018

Stanford University Stanford, CA
Research Assistant at Department of Statistics Jan 2017 – Jun 2017

The World Bank Group Washington, DC
Short Term Temporary at Department of Trade and Competitiveness Jun 2016 – Aug 2016

China Investment Corporation Beijing, China
Analyst Intern at Special Investment Department Jul 2013 – Aug 2013

Research Interests

Integrated assessment modeling
Air pollution and health impacts
Health inequity and environmental justice

Realistic modeling of decarbonization pathways

Peer-reviewed Publications

Xinyuan Huang, Jinyu Shiwang, Klaus Keller, Wei Peng. "Fine-scale population, emissions, and atmospheric dynamics drive health disparity outcomes from energy transition." *Earth's Future* (2026)

Jinyu Shiwang, Wei Peng, **Xinyuan Huang**, Gokul Iyer, Vivek Srikrishnan, Klaus Keller. "Technology indicators enable early detection of decarbonization failures." Applied Energy (2026)

Xinyuan Huang, Wei Peng, Alicia Zhao, Yang Ou, Shannon Kennedy, Gokul Iyer, Haewon McJeon, Ryna Cui, Nate Hultman. "Substantial air quality and health co-benefits from combined federal and subnational climate actions in the United States." *One Earth* (2025)

Wei Peng, Susan Anenberg, John Bistline, …, **Xinyuan Huang**, et al. "Seizing the policy opportunities for health- and equity-improving energy decisions." *One Earth* (2025)

Xinyuan Huang, Vivek Srikrishnan, Jonathan Lamontagne, Klaus Keller, and Wei Peng. "Effects of global climate mitigation on regional air quality and health." *Nature Sustainability* (2023)

Hui Yang, **Xinyuan Huang**, Dan Westervelt, Larry Horowitz, Wei Peng. "Socio-demographic factors shaping the future global health burden from air pollution." *Nature Sustainability* (2023)

Working Paper and In-Review

Wei Peng*, Aniruddha Deshpande*, **Xinyuan Huang***, Jinyu Shiwang*, Frank Errickson, Gokul Iyer, Daniel H. Loughlin, Christopher G. Nolte, Steven J. Smith, Fabian Wagner, Pengfei Wang, Jie Zhang, Mark Budolfson, Noah Scovronick. "Complex factors influencing US air pollution equity under a Net-Zero transition." (Revise and resubmit at *Nature Sustainabilit*y)
*These authors contributed equally to this work.

Carla Campos, Yueqi Jiang, Kerem Ziya Akdemir, **Xinyuan Huang**, Dan Loughlin, Kendall Mongrid, Jennie Rice, Nathalie Voisin, Wei Peng. "Impacts of fine-scale power system dynamics on future air quality: A test case on the Western US." (To be submitted)

Jinyu Shiwang, Wei Peng, John Bistline, Jamil Farbes, **Xinyuan Huang**, Gokul Iyer, Jesse D. Jenkins, Eladio Knipping, Nicholas Mailloux, Erin Mayfield, Aranya Venkatesh, Qianru Zhu. "Energy technology choices shaping the air quality and health effects of a net-zero transition". (Revise and resubmit at *Environmental Science and Technology*)

Other Publications

Xinyuan Huang, Alicia Zhao, Ryna Cui, Wei Peng. "Bridging climate and health: A collaborative path to policy-relevant science." *iScience* (2025)

Alicia Zhao, Ryna Cui, Carla Campos Morales, Jiaxun Sun, Wei Peng, Claire Squire, Kiara Ordonez Olazabal, Xiangjie Chen, **Xinyuan Huang**, Kuishuang Feng, Adriana Bryant, Laura Hinkle, Stephanie Vo, Michelle Faggert, Camryn Dahl, and Nathan Hultman. *U.S. Clean Energy Policy Rollbacks: The Economic and Public Health Impacts Across States.* Policy report. Center for Global Sustainability, University of Maryland. June 2025

Wei Peng, **Xinyuan Huang**, Alicia Zhao, Yang Ou, Shannon Kennedy, Gokul Iyer, Haewon McJeon, Ryna Cui, Nate Hultman. *All-In Climate Action for Improved US Air Quality and Health Benefits*. Policy report. Center for Global Sustainability, University of Maryland and America is All In. December 2023

Teaching Experience

Teaching Assistant
Water and wastewater treatment (CE 371), Penn State University — Fall 2019, Fall 2021

Guest Lectures
Energy-Environment Nexus (CE 597), Penn State University — Fall 2023

Conference Presentations

2025
Princeton E-ffiliates Partnership Retreat (Poster) — Princeton, NJ

2024
AGU Annual Meeting (Oral and poster) — Washington, DC
GCAM Annual Meeting (Poster) — College Park, MD
American Meteorological Society (AMS) Annual Meeting (Oral) — Baltimore, MD

2023
AGU Annual Meeting (Poster) — San Francisco, CA

2022
AGU Fall Meeting (Poster) — Chicago, IL
Integrated Assessment Modeling Consortium (IAMC) Annual Meeting (Oral) — College Park, MD
INFORMS Annual Meeting (Oral) — Indianapolis, IN

2021
AGU Fall Meeting (Poster) — New Orleans, LA

Invited Talks

2024
GCAM-USA State Applications Community of Practice
Andlinger Center for Energy and the Environment 2024 Summer Seminar

2023
The Joint Global Change Research Institute

Service

- Member of American Geophysical Union (AGU), American Meteorological Society (AMS)
- Reviewer: Communications Earth and Environment, Energy and Climate Change, Environmental Science and Technology, Frontiers in Public Health

Skills

- Languages: English (professional proficiency), Mandarin Chinese (native)
- Programming and Computing: R, Python, SQL, Linux, HPC/cluster computing

- Modeling and Analytical Tools: Global Change Analysis Model (GCAM), International Futures (IFs), Intervention Model for Air Pollution (InMAP), The Environmental Benefits Mapping and Analysis Program (BenMAP)